



From spreadsheets to SQL to Python

Although the Python programming language might be new to you, it actually has a lot of similarities to the other tools you have explored in this program. In this reading, you will compare spreadsheet programs, SQL, and Python to have a better sense of how to use each tool moving forward.



Spreadsheets, SQL, and Python Comparison

As a data analyst, there is a good chance you will work with spreadsheet programs, SQL, and Python at some point in your career. Each tool has its own strengths and weaknesses, but they all make the data analysis process smoother and more efficient. There are two main things that all three tools have in common:

- They all use filters: For example, you can easily filter a dataset using any of these tools. In Python, you can use the Pandas library to filter data. In SQL, you can use a basic `SELECT-FROM-WHERE` query. In a spreadsheet, you can create a filter using the menu options.
- They all use functions: In spreadsheets, you use functions in formulas, and in SQL, you include them in queries. In Python, you will use functions in the code that is part of your analysis.

The table below presents key questions to further explore how these tools compare to each other. You can use this as a general guideline as you begin to navigate Python.

Key question	Spreadsheet Programs	SQL (Structured Query Language)	Python programming language

What is it?	A program that uses rows and columns to organize data and allows for analysis and manipulation through formulas, functions, and built-in features	A database programming language used to communicate with databases to conduct an analysis of data	A general purpose programming language used for data analysis, scientific computing, machine learning, web development, and more
What is a primary advantage?	Includes a variety of visualization tools/features	Allows users to manipulate and reorganize data as needed to aid analysis	Provides an accessible language with extensive libraries (like Pandas and NumPy) for data manipulation, analysis, and visualization
Which datasets does it work best with?	Smaller datasets	Larger datasets	Larger datasets

What is the source of the data?	Entered manually or imported from an external source	Accessed from an external database	Can be loaded from various sources like CSV files, spreadsheets, databases, or APIs
Where is the data from my analysis usually stored?	In a spreadsheet file on your computer	Inside tables in the accessed database	In a Python file or Jupyter Notebook on your computer
Do I use formulas and functions?	Yes	Yes	Yes
Can I create visualizations?	Yes	Yes, by using an additional tool like a database management system (DBMS) or a business intelligence (BI) tool	Yes, using libraries such as Seaborn, Matplotlib, or Plotly

One of the exciting things about all of the tools you've been learning is that they are often used together. For example, you could use SQL to pull a specific dataset, clean and organize it in Python, and then export it as a spreadsheet for quick insights. As you learn more about Python, you may discover even more exciting ways it can be combined with other tools in your toolkit!